

HW 10

1. Let X, Y, Z be discrete RVs w/ joint prob. $p_{ijk}^{xyz} = \text{Prob}(X=x_i, Y=y_j, Z=z_k)$

$$x_1 = -4 \quad x_2 = 2 \quad x_3 = 4 \quad x_4 = 8$$

$$y_1 = 0 \quad y_2 = 3 \quad y_3 = 6$$

$$z_1 = 2 \quad z_2 = -4$$

$$a) p_{ij}^{xy} = \sum_{k=1}^2 p_{ijk}^{xyz}$$

$$p_{jk}^{yz} = \sum_{i=1}^4 p_{ijk}^{xyz}$$

$$p_{ik}^{xz} = \sum_{j=1}^3 p_{ijk}^{xyz}$$

$j \backslash i$	1	2	3	4
1	0.15	0.04	0.04	0.07
2	0.08	0.12	0.12	0.08
3	0.02	0.04	0.04	0.20

$k \backslash j$	1	2	3
1	0.23	0.33	0.19
2	0.07	0.07	0.11

$k \backslash i$	1	2	3	4
1	0.22	0.17	0.17	0.19
2	0.03	0.03	0.03	0.16

$$b) p_i^x = \sum_{j=1}^3 p_{ij}^{xy}$$

$i \backslash j$	1	2	3	4
p_i^x	0.25	0.20	0.20	0.35

$$p_j^y = \sum_{k=1}^2 p_{jk}^{yz}$$

$j \backslash k$	1	2	3
p_j^y	0.30	0.40	0.30

$$p_k^z = \sum_{i=1}^4 p_{ik}^{xz}$$

$k \backslash i$	1	2
p_k^z	0.75	0.25

$$c) \Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) & \text{Cov}(X, Z) \\ \text{Cov}(Y, X) & \text{Var}(Y) & \text{Cov}(Y, Z) \\ \text{Cov}(Z, X) & \text{Cov}(Z, Y) & \text{Var}(Z) \end{pmatrix} = \begin{pmatrix} 21.4 & 4.68 & -3.54 \\ 4.68 & 5.4 & -0.72 \\ -3.54 & -0.72 & 6.5 \end{pmatrix}$$

$E[\cdot]$

$$E[X] = (-4)(0.25) + 2(0.2) + 4(0.2) + 8(0.35) = 3$$

$$E[Y] = 0(0.3) + 3(0.4) + 6(0.3) = 3$$

$$E[Z] = 2(0.75) + (-4)(0.25) = 0.5$$

$\text{Cov}(\cdot, \cdot)$

$$\text{Cov}(X, Y) = 13.68 - (3)(3) = 4.68$$

$$\hookrightarrow E[XY] = 13.68 \text{ (using online calc)}$$

$$\text{Cov}(X, Z) = -2.04 - (3)(0.5) = -3.54$$

$$\hookrightarrow E[XZ] = -2.04 \text{ (using online calc)}$$

$$\text{Cov}(Y, Z) = 0.78 - (3)(0.5) = -0.72$$

$$\hookrightarrow E[YZ] = 0.78 \text{ (using online calc)}$$

$\text{Var}(\cdot)$

$$\text{Var}(X) = 30.4 - 9 = 21.4$$

$$\hookrightarrow E[X^2] = 16(0.25) + 4(0.2) + 16(0.2) + 64(0.35) = 30.4$$

$$\text{Var}(Y) = 14.4 - 9 = 5.4$$

$$\hookrightarrow E[Y^2] = 0(0.3) + 9(0.4) + 36(0.3) = 14.4$$

$$\text{Var}(Z) = 7 - 0.5 = 6.5$$

$$\hookrightarrow E[Z^2] = 4(0.75) + 16(0.25) = 7$$

2. Let X, Y be RVs. Show that $\text{Var}(X+Y) = \text{Var}(X) + 2\text{Cov}(X, Y) + \text{Var}(Y)$

$$\begin{aligned}\text{Var}(X+Y) &= E[(X+Y)^2] - E[X+Y]^2 = E[X^2 + 2XY + Y^2] - E[X+Y]^2 && (x+y)^2 = x^2 + 2xy + y^2 \\ &= E[X^2] + E[2XY] + E[Y^2] - (E[X]^2 + 2E[X]E[Y] + E[Y]^2) && (E[X] + E[Y])^2 \\ &= (E[X^2] - E[X]^2) + (2E[XY] - 2E[X]E[Y]) + (E[Y^2] - E[Y]^2) \\ &= \text{Var}(X) + 2\text{Cov}(X, Y) + \text{Var}(Y) \quad \checkmark\end{aligned}$$

If X and Y are independent, then by definition, the covariance between X and Y is 0. This is due to the fact that $E[XY] = E[X]E[Y]$, given that X and Y has no linear relationship, so those terms cancel, leaving us with

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) \quad \checkmark$$

3. Let $x_1, x_2, \dots, x_n$ be n samples of RV X

$$a) E[u] = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \frac{1}{n} E\left[\sum_{i=1}^n x_i\right] = \frac{1}{n} \sum_{i=1}^n E[x_i] = \frac{n}{n} E[X] = E[X] \quad \checkmark$$

$$b) S = \frac{1}{n-1} \sum_{i=1}^n (x_i - u)^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - 2 \sum_{i=1}^n x_i u + \sum_{i=1}^n u^2 \right) = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - 2u \sum_{i=1}^n x_i + n u^2 \right)$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - 2n u^2 + n u^2 \right) = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n u^2 \right)$$

$$E[S] = E\left[\frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n u^2 \right)\right] \stackrel{i.i.d}{=} \frac{1}{n-1} E\left[\sum_{i=1}^n x_i^2 - n u^2\right] = \frac{1}{n-1} \left(\sum E[x_i^2] - n E[u^2] \right) = \frac{1}{n-1} \left(n \text{Var}(X) + n u^2 - n \left(\frac{\text{Var}(X)}{n} + u^2 \right) \right)$$

$$\text{Var}(X) = E[X^2] - u^2 \Rightarrow E[X^2] = \text{Var}(X) + u^2$$

$$\hookrightarrow E[u^2] = \text{Var}(u) + E[u]^2$$

$$= \frac{1}{n-1} \left(n \text{Var}(X) + \cancel{n u^2} - \text{Var}(X) - \cancel{n u^2} \right) = \frac{1}{n-1} (n-1) \text{Var}(X)$$

$$= \frac{\text{Var}(X)}{n} + u^2$$

$$= \text{Var}(X) \quad \checkmark$$

4. random vector $v \in \mathbb{R}^n$ is Gaussian if $v_1, \dots, v_n \sim N(0, 1)$

$$E[v] = E[a_1 v_1 + \dots + a_n v_n] = \sum_{i=1}^n a_i E[v_i] = \sum_{i=1}^n a_i \cdot \vec{0} = \vec{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\text{Var}(v) = I$$

$$\Sigma = E[(v - E[v])(v - E[v])^T] = E[vv^T]$$

$$\hookrightarrow \Sigma_{ij} = \text{Cov}(v_i, v_j) = \begin{cases} \text{Var}(v_i) = 1 & \text{if } i=j \\ \text{Cov}(v_i, v_j) = 0 & \text{if } i \neq j \end{cases}$$

$$\Sigma = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & \\ \vdots & & \ddots & \\ 0 & 0 & \dots & 1 \end{bmatrix} = I_n$$

5. a) in code below

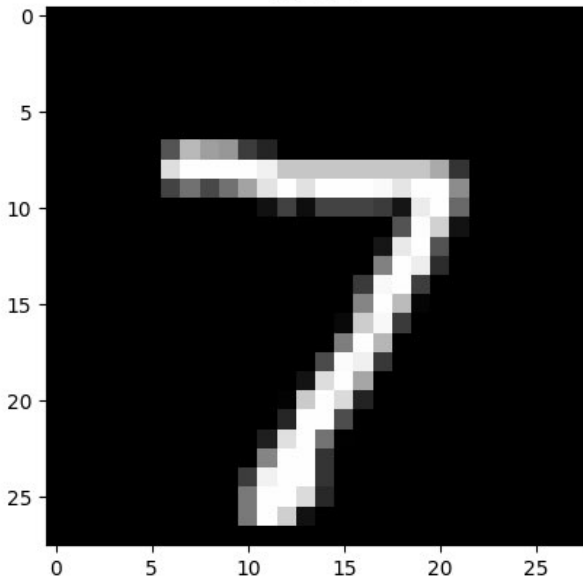
first image: 7

b) in code below

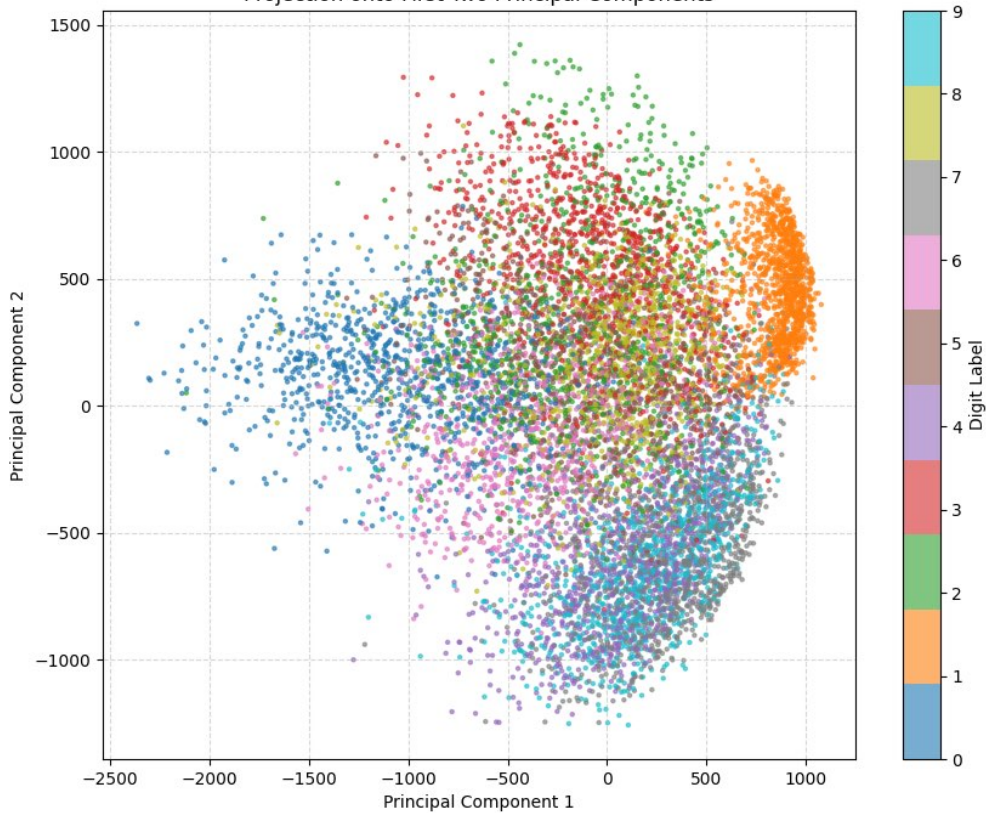
c) While PCA is a linear technique, the first two principal components still capture a significant amount of the variance. Certain digits such as 0 and 1 usually form very distinct clusters due to their shapes being geometrically unique.

To label a new digit vector $v \in \mathbb{R}^{28 \times 9}$ using only PCA results, we would subtract the original training mean from the new point $v_{\text{centered}} = v - u_{\text{train}}$, project this into 2D subspace, and get a single point (z_1, z_2) in $\mathbb{R}^2$. We then compare this point to the coordinates of the known data points in the scatter plot, assigning the label of the closest known point or cluster centroid.

Label: 7



Projection onto First Two Principal Components



```

import numpy as np
import matplotlib.pyplot as plt
from scipy.io import loadmat

# Load the data
data = loadmat('hw/mnist.mat')
testX = data['testX']
testY = data['testY']

# Extract first image in X and reshape to 28x28
first_image_flat = testX[0, :]

X_image = first_image_flat.reshape(28, 28).astype(float)

# 3. Visualize
plt.imshow(X_image, cmap='gray')
plt.title(f"Label: {testY[0][0]}")
plt.show()

print(f"First entry of testY: {testY[0][0]}")

# i. Convert to double precision
A = testX.astype(np.float64)

# ii. Center the data
column_means = np.mean(A, axis=0)
A_centered = A - column_means # A - mu

# iii. Compute SVD
U, S, Vt = np.linalg.svd(A_centered, full_matrices=False)

# c) Project first two principal components
PCs = Vt[:2, :].T
X_projected = A_centered @ PCs

plt.figure(figsize=(10, 8))
scatter = plt.scatter(X_projected[:, 0], X_projected[:, 1],
c=testY.flatten(
), cmap='tab10', s=5, alpha=0.6)
plt.colorbar(scatter, ticks=range(10), label='Digit Label')
plt.title("Projection onto First Two Principal Components")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()

'''TERMINAL OUTPUT
First entry of testY: 7
'''

```